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Part # / Keyword



0

[Home](#) > [NEWEST Products](#) > [New by Manufacturer](#) > [Arduino](#) > [Modulino® Knob - Arduino](#)

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[← Back to Arduino](#)

Arduino Modulino® Knob

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Arduino Modulino® Knob is a compact, plug-and-play input module that features a quadrature rotary encoder (PEC11J-9215F-S0015) paired with an onboard STM32C011F4 microcontroller, enabling both precise rotational input and push-button interaction. The module is designed to integrate seamlessly with the Arduino ecosystem. Featuring a high-quality rotary encoder with a tactile push-button, the Modulino Knob allows users to implement intuitive control interfaces for a wide range of applications, such as menu navigation, parameter adjustment, and interactive projects. It connects via the Qwiic I²C connector, enabling easy daisy-chaining with other Modulino components without soldering. The module also includes a built-in RGB LED for visual feedback, enhancing user interaction. With a robust design and

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Arduino's signature ease of use, the Modulino Knob is ideal for both prototyping and educational environments.

FEATURES

- PEC11J-9215F-S0015 quadrature rotary encoder with push-switch capability for rich user input
- STM32C011F4 microcontroller handling I²C communications and optional reprogramming
- Operates at 3.3V via the Qwiic interface, supports 2.0V to 3.6V supply range internally
- SWD interface onboard for firmware customization, plus optional UART lines
- Plug-and-play connectivity
- Robust, ready-to-use design
- Effortless expansion
- Seamless Arduino ecosystem integration
- Built-in RGB LED
- I²C Qwiic cable included
- CE/RED (Europe), UKCA (UK), FCC (USA), IC (Canada), RoHS, REACH, and WEEE certified
- Ideal for menu navigation, fine-tuning controls, and interactive IoT or maker projects

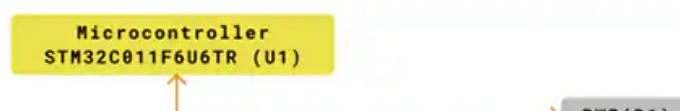
APPLICATIONS

- User interface controls
- Industrial/mechanical projects
- Educational tools

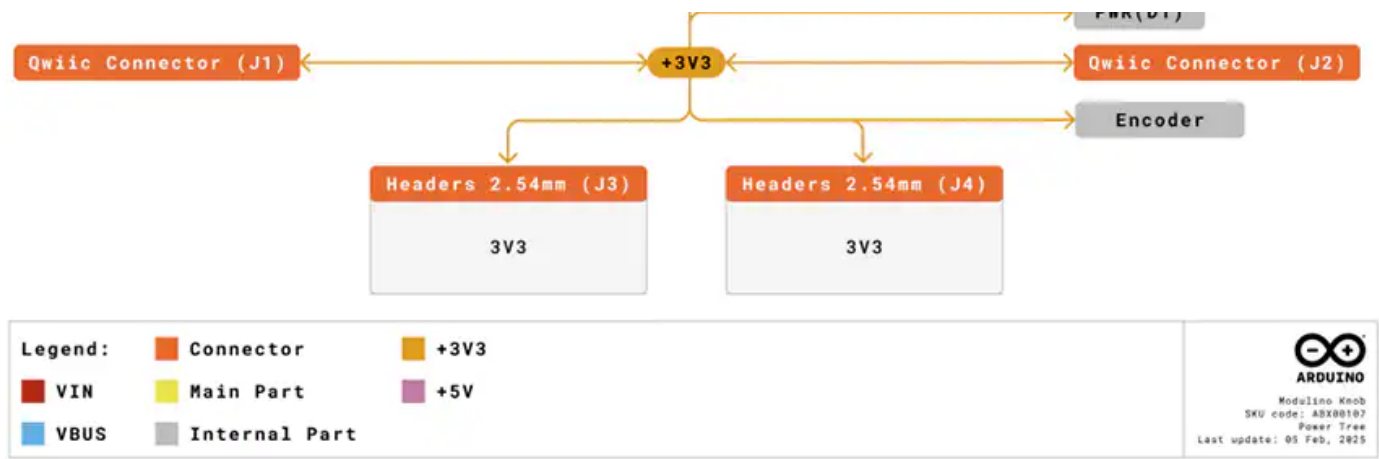
SPECIFICATIONS

- STM32C011F4 MCU, handles I²C and rotary encoder logic, 2.0V to 3.6V MCU supply range
- PEC11J-9215F-S0015 encoder (quadrature + push switch), 360° encoder range (continuous rotation)
- I²C (Qwiic), SWD (debug/reprogram), and optional UART communication
- 3.3V nominal operating voltage via Qwiic
- ~3.4mA typical current consumption (microcontroller + rotary encoder interface)
- Board dimensions
 - 41mm x 25.36mm size
 - 1.6mm (±0.2mm) thickness
 - 4x ø3.2mm mounting holes
 - 16mm vertical, 32mm horizontal hole spacing
- -40°C to +85°C operating temperature range

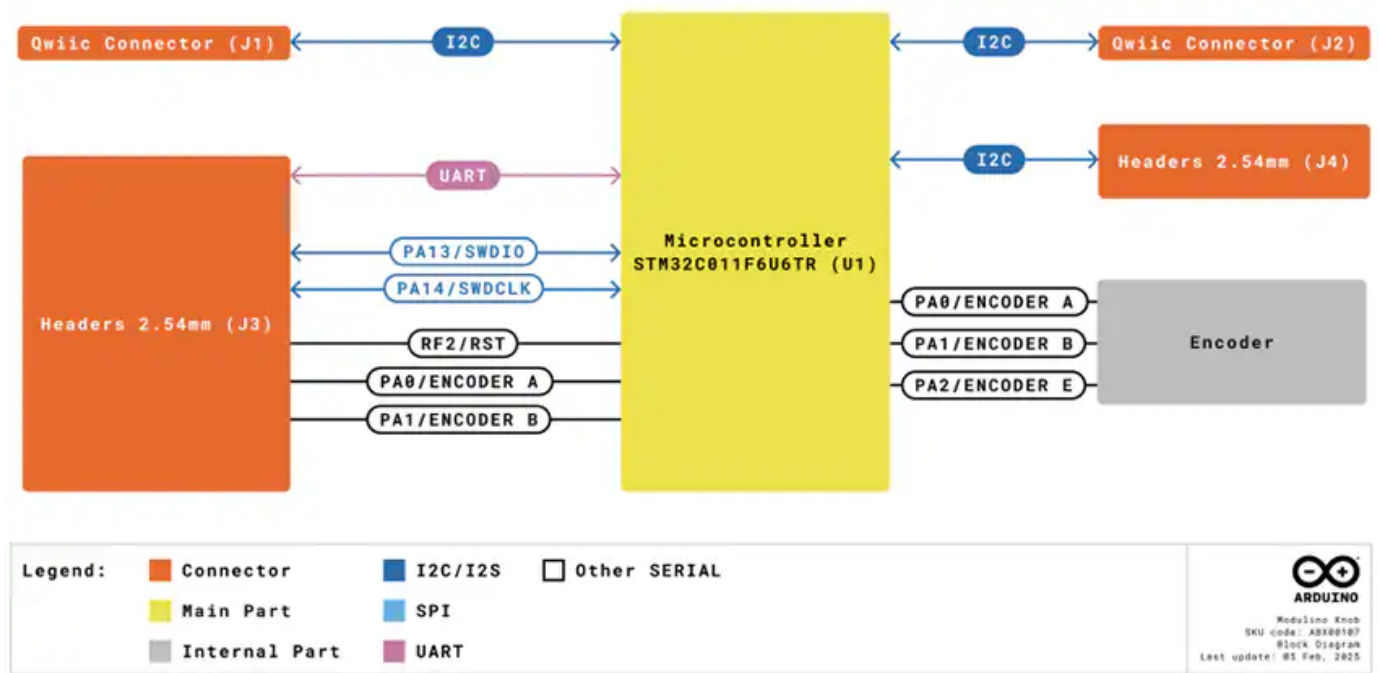
POWER TREE



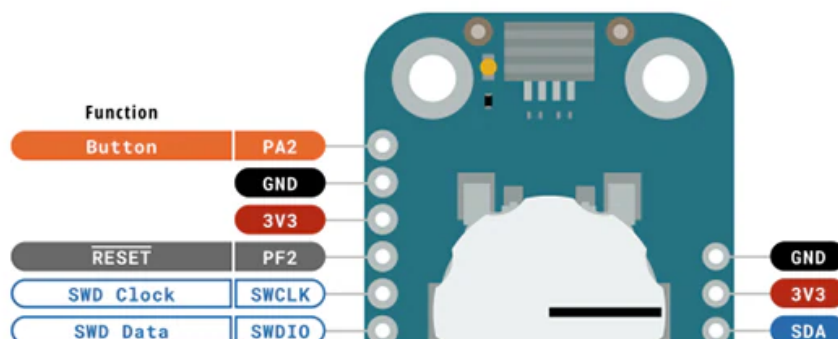
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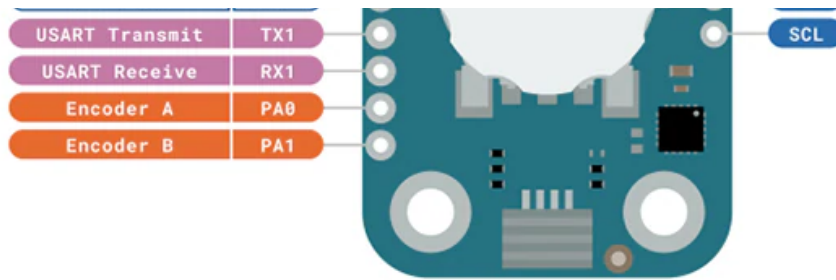


BLOCK DIAGRAM



PINOUT

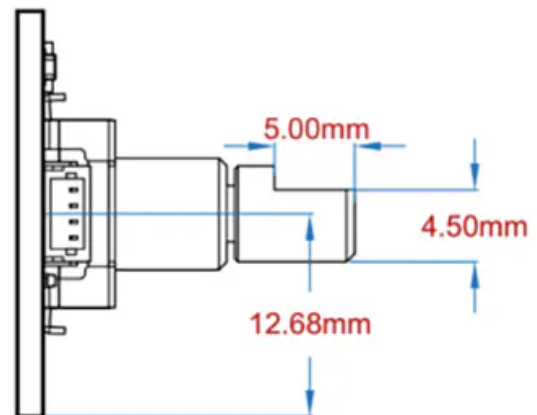
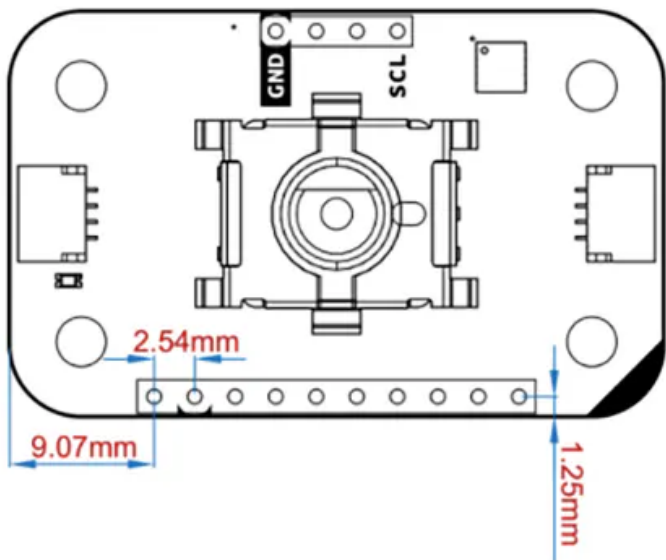
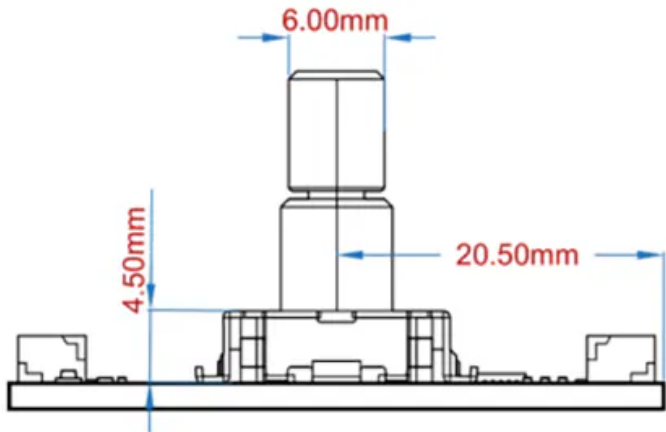
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Legend:	Digital	I2C	Other SERIAL
	Power	Analog	Analog
	Ground	Main Part	UART/USART
		SPI	PWM/Timer

Modulino Knob
SKU code: ABX00107
Pinout
Last update: 18 Jun, 2024

DIMENSIONS



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